CMSC 5718 Assignment 1: Derivative Pricing and Hedging

Group Members:

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Note: We have chosen Chen Guanyu's student number, which ends in 5, as the selection number for this assignment.

Student number last digit: 5

Parameter	Value
S_1	China Construction Bank (939), $S_0 = 7.92$, $\sigma = 0.202$
S_2	Agricultural Bank (1288), $S_0 = 5.60$, $\sigma = 0.232$
Correlation ρ	0.621
Number of options M	250,000
Risk-free rate r	3.14% p.a. (continuously compounded)

Note: Dividends, transaction costs and board lots are ignored.

In [1]:
```
import numpy as np
from scipy.stats import norm
from scipy.optimize import brentq
import time
import pandas as pd
from datetime import datetime

np.random.seed(42)

# ============================================================================
# Common Parameters
# ============================================================================
r = 0.0314          # continuously compounded interest rate
S1_0 = 7.92         # initial price of S1 (CCB 939) as of June 30, 2025
S2_0 = 5.60         # initial price of S2 (Agricultural Bank 1288)
sigma1 = 0.202      # volatility of S1
sigma2 = 0.232      # volatility of S2
rho = 0.621         # correlation coefficient
M = 250000          # number of options
K_option = S1_0     # ATM strike price (S = K = 7.92)

print(f"S1: China Construction Bank (939), S0 = {S1_0}, σ = {sigma1}")
print(f"S2: Agricultural Bank (1288), S0 = {S2_0}, σ = {sigma2}")
```

```
print(f"Correlation ρ = {rho}")
print(f"Risk-free rate r = {r}")
print(f"Number of options M = {M:,}")
```

```
S1: China Construction Bank (939), S0 = 7.92, σ = 0.202
S2: Agricultural Bank (1288), S0 = 5.6, σ = 0.232
Correlation ρ = 0.621
Risk-free rate r = 0.0314
Number of options M = 250,000
```

Question 1: European Call Option Pricing (20%)

Parameters:

- European call option, at-the-money (S = K = 7.92)
- r = 3.14% p.a., T = 0.5014 year
- σ = 0.202

Q1(i): Black-Scholes Analytical Price

In [2]:
```
# Q1(i): Black-Scholes Analytical Price
T1 = 0.5014  # maturity
S = S1_0
K = K_option  # ATM: S = K = 7.92

d1 = (np.log(S / K) + (r + 0.5 * sigma1**2) * T1) / (sigma1 * np.sqrt(T1))
d2 = d1 - sigma1 * np.sqrt(T1)

bs_price = S * norm.cdf(d1) - K * np.exp(-r * T1) * norm.cdf(d2)

print(f"Black-Scholes European Call Option Pricing")
print(f"{'='*50}")
print(f"  S = K = {K}")
print(f"  T = {T1} years")
print(f"  r = {r}")
print(f"  σ = {sigma1}")
print(f"{'-'*50}")
print(f"  d1    = {d1:.6f}")
print(f"  d2    = {d2:.6f}")
print(f"  N(d1) = {norm.cdf(d1):.6f}")
print(f"  N(d2) = {norm.cdf(d2):.6f}")
print(f"{'-'*50}")
print(f"  ★ BS Call Price = {bs_price:.6f}")
print(f"{'='*50}")
```

```
Black-Scholes European Call Option Pricing
==================================================
  S = K = 7.92
  T = 0.5014 years
  r = 0.0314
  σ = 0.202
--------------------------------------------------
  d1    = 0.181588
  d2    = 0.038553
  N(d1) = 0.572047
  N(d2) = 0.515376
--------------------------------------------------
  ★ BS Call Price = 0.512591
==================================================
```

Q1(ii): Monte Carlo Pricing

Using Euler discretization of GBM with N = 120 time steps, Δt = T/N = 0.004178.

$$S(t+\Delta t) = S(t) \cdot \exp\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} \cdot Z\right], \quad Z \sim N(0,1)$$

In [3]:

```python
# Q1(ii): Monte Carlo Pricing
N_steps = 120
dt = T1 / N_steps

def monte_carlo_european_call(S0, K, r, sigma, T, N_steps, n_paths):
    """Monte Carlo pricing using Euler discretization of GBM."""
    dt = T / N_steps
    S = np.full(n_paths, S0, dtype=np.float64)
    for _ in range(N_steps):
        Z = np.random.standard_normal(n_paths)
        S = S * np.exp((r - 0.5 * sigma**2) * dt + sigma * np.sqrt(dt) * Z)
    payoff = np.maximum(S - K, 0)
    price = np.exp(-r * T) * np.mean(payoff)
    std_err = np.exp(-r * T) * np.std(payoff) / np.sqrt(n_paths)
    return price, std_err

# (a) 10,000 paths
np.random.seed(42)
start_time = time.time()
mc_price_10k, mc_se_10k = monte_carlo_european_call(S, K, r, sigma1, T1, N_s
time_10k = time.time() - start_time

# (b) 500,000 paths
np.random.seed(42)
start_time = time.time()
mc_price_500k, mc_se_500k = monte_carlo_european_call(S, K, r, sigma1, T1, N
time_500k = time.time() - start_time

# Display results as a table
results_q1 = pd.DataFrame({
    'Method': ['Black-Scholes', 'MC (10,000 paths)', 'MC (500,000 paths)'],
    'Price': [bs_price, mc_price_10k, mc_price_500k],
```

```
    'Std Error': ['-', f'{mc_se_10k:.6f}', f'{mc_se_500k:.6f}'],
    'Time (s)': ['-', f'{time_10k:.4f}', f'{time_500k:.4f}'],
    'Diff from BS': ['-', f'{abs(mc_price_10k - bs_price):.6f}', f'{abs(mc_p
})
print(f"N = {N_steps} time steps, Δt = {dt:.6f}")
print()
display(results_q1.style.set_caption("Q1: European Call Option Pricing Resul
```

```
N = 120 time steps, Δt = 0.004178
```

Q1: European Call Option Pricing Results

Method	Price	Std Error	Time (s)	Diff from BS
Black-Scholes	0.512591	-	-	-
MC (10,000 paths)	0.505949	0.007441	0.0262	0.006642
MC (500,000 paths)	0.511162	0.001072	0.9392	0.001429

Question 2: Exotic Option Pricing (35%)

An exotic option based on two stocks S_1 and S_2 with T = 0.5 year:

$$B = \min\left(\frac{S_{1,1}}{S_{1,0}}, \frac{S_{2,1}}{S_{2,0}}\right)$$

$$\text{Payoff} = \begin{cases} B - 1, & \text{if } B < 1 \text{ or } B > 1 + K \\ K, & \text{if } 1 \leq B \leq 1 + K \end{cases}$$

Goal: Find K such that the fair price = −0.03.

Using Monte Carlo with **one time step** and **1,000,000 paths**. Correlated normals via Cholesky decomposition.

In [4]:
```
# Q2: Exotic Option Pricing
T2 = 0.5  # maturity for exotic option
n_paths_q2 = 1000000

def exotic_option_price(K_exotic, S1_0, S2_0, sigma1, sigma2, rho, r, T, n_p
    """
    Price the exotic option using Monte Carlo with one time step.
    B = min(S1_T/S1_0, S2_T/S2_0)
    Payoff = B - 1,  if B < 1 or B > 1 + K_exotic
           = K_exotic, if 1 <= B <= 1 + K_exotic
    """
    # Generate correlated normals using Cholesky decomposition
    Z1 = np.random.standard_normal(n_paths)
    Z2 = np.random.standard_normal(n_paths)
    W1 = Z1
    W2 = rho * Z1 + np.sqrt(1 - rho**2) * Z2
```

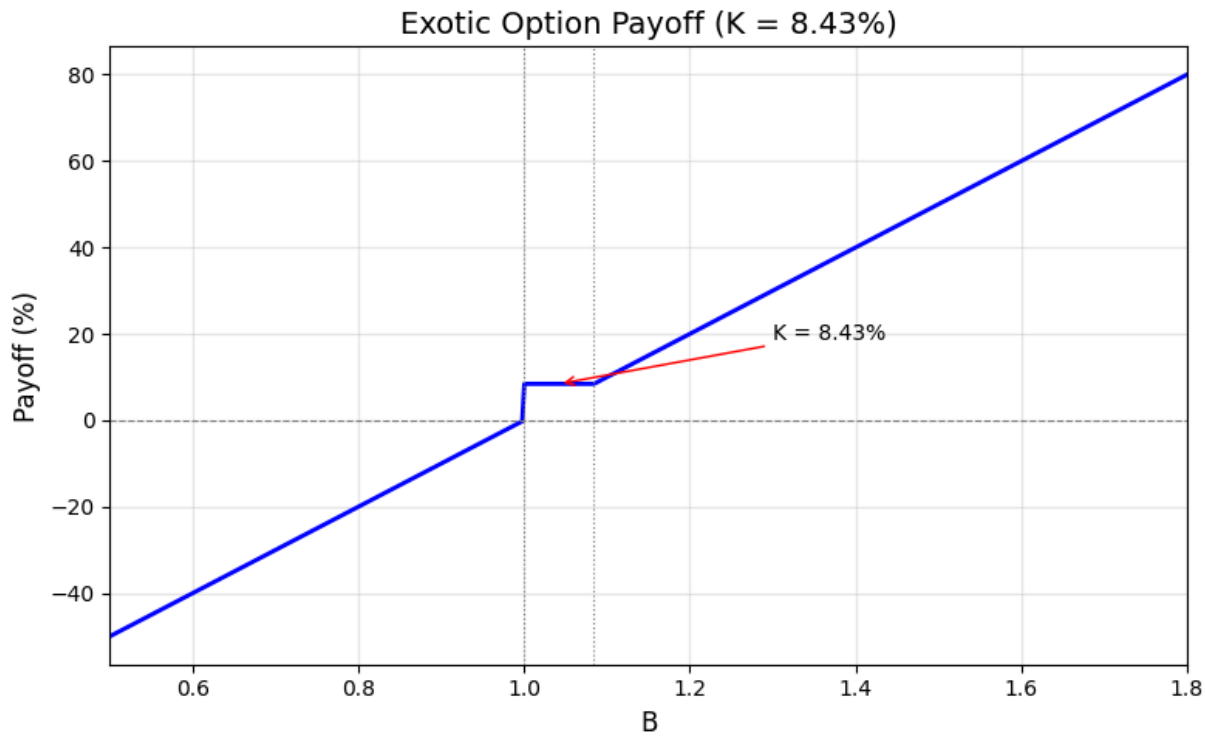
```
    # Simulate terminal stock prices (one time step)
    S1_T = S1_0 * np.exp((r - 0.5 * sigma1**2) * T + sigma1 * np.sqrt(T) * W
    S2_T = S2_0 * np.exp((r - 0.5 * sigma2**2) * T + sigma2 * np.sqrt(T) * W

    # Calculate B
    B = np.minimum(S1_T / S1_0, S2_T / S2_0)

    # Calculate payoff
    payoff = np.where(
        (B >= 1) & (B <= 1 + K_exotic),
        K_exotic,
        B - 1
    )

    # Discounted price
    price = np.exp(-r * T) * np.mean(payoff)
    return price

def price_minus_target(K_exotic, target=-0.03):
    """Function for root finding: price(K) - target = 0"""
    np.random.seed(123)
    price = exotic_option_price(K_exotic, S1_0, S2_0, sigma1, sigma2, rho, r
    return price - target

# Scan to find approximate range
print("Scanning K values to find the root bracket...")
K_test_values = np.arange(0.01, 1.0, 0.02)
prices_test = []
for k_val in K_test_values:
    np.random.seed(123)
    p = exotic_option_price(k_val, S1_0, S2_0, sigma1, sigma2, rho, r, T2, r
    prices_test.append(p)

# Find sign change
K_low, K_high = None, None
for i in range(len(prices_test) - 1):
    if (prices_test[i] - (-0.03)) * (prices_test[i + 1] - (-0.03)) < 0:
        K_low, K_high = K_test_values[i], K_test_values[i + 1]
        print(f"  Root bracket found: K ∈ [{K_low:.4f}, {K_high:.4f}]")
        break

# Refine using Brent's method
K_solution = brentq(price_minus_target, K_low, K_high, xtol=1e-6)

# Verify
np.random.seed(123)
final_price_q2 = exotic_option_price(K_solution, S1_0, S2_0, sigma1, sigma2,

print(f"\n{'='*50}")
print(f"  ★ K = {K_solution:.6f} ({K_solution*100:.4f}%)")
print(f"  Fair price at this K = {final_price_q2:.6f}")
print(f"  Target price         = -0.030000")
print(f"  Difference           = {abs(final_price_q2 - (-0.03)):.8f}")
print(f"{'='*50}")
```

```
Scanning K values to find the root bracket...
  Root bracket found: K ∈ [0.0700, 0.0900]

==================================================
  ⋆ K = 0.084339 (8.4339%)
  Fair price at this K = -0.030000
  Target price         = -0.030000
  Difference           = 0.00000000
==================================================
```

Payoff Diagram for the Exotic Option

In [5]:

```python
import matplotlib.pyplot as plt

# Plot the payoff diagram
B_vals = np.linspace(0.5, 1.8, 500)
K_plot = K_solution

payoff_vals = np.where(
    (B_vals >= 1) & (B_vals <= 1 + K_plot),
    K_plot,
    B_vals - 1
)

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(B_vals, payoff_vals * 100, 'b-', linewidth=2)
ax.axhline(y=0, color='gray', linestyle='--', linewidth=0.8)
ax.axvline(x=1, color='gray', linestyle=':', linewidth=0.8)
ax.axvline(x=1 + K_plot, color='gray', linestyle=':', linewidth=0.8)
ax.set_xlabel('B', fontsize=12)
ax.set_ylabel('Payoff (%)', fontsize=12)
ax.set_title(f'Exotic Option Payoff (K = {K_plot*100:.2f}%)', fontsize=14)
ax.set_xlim(0.5, 1.8)
ax.grid(True, alpha=0.3)

# Annotate key points
ax.annotate(f'K = {K_plot*100:.2f}%', xy=(1 + K_plot/2, K_plot*100),
            xytext=(1.3, K_plot*100 + 10), fontsize=10,
            arrowprops=dict(arrowstyle='->', color='red'))

plt.tight_layout()
plt.show()
```



Question 3: Delta Hedging Strategy (45%)

Q3(i): Construct Delta Hedging Table

Setup:

- Short M = 250,000 call options on S_1 (CCB 939) as of June 30, 2025
- Strike K = 7.92 (ATM), σ = 0.202, r = 3.14%
- Hedging period: June 30, 2025 → December 30, 2025
- Delta neutral daily rebalancing

Daily account balance:

$$\text{Balance}_t = \text{Balance}_{t-1} + \text{Interest}_t - \text{Cost of shares traded}_t$$

$$\text{Interest}_t = \text{Balance}_{t-1} \times \left[e^{r \times \frac{\text{num_days}}{365}} - 1 \right]$$

In [6]:
```
# Q3(i): Delta Hedging Strategy
# Daily price data for stock 939 (CCB) from June 30 to Dec 30, 2025
price_data = [
    ("2025-06-30", 7.92), ("2025-07-02", 8.15), ("2025-07-03", 8.15), ("2025
    ("2025-07-07", 8.20), ("2025-07-08", 8.24), ("2025-07-09", 8.22), ("2025
    ("2025-07-11", 8.35), ("2025-07-14", 8.41), ("2025-07-15", 8.42), ("2025
    ("2025-07-17", 8.31), ("2025-07-18", 8.47), ("2025-07-21", 8.37), ("2025
    ("2025-07-23", 8.38), ("2025-07-24", 8.42), ("2025-07-25", 8.30), ("2025
    ("2025-07-29", 8.19), ("2025-07-30", 8.19), ("2025-07-31", 8.05), ("2025
```

```
    ("2025-08-04", 7.92), ("2025-08-05", 8.00), ("2025-08-06", 7.93), ("2025
    ("2025-08-08", 7.94), ("2025-08-11", 7.93), ("2025-08-12", 7.95), ("2025
    ("2025-08-14", 7.98), ("2025-08-15", 7.80), ("2025-08-18", 7.71), ("2025
    ("2025-08-20", 7.75), ("2025-08-21", 7.74), ("2025-08-22", 7.75), ("2025
    ("2025-08-26", 7.55), ("2025-08-27", 7.49), ("2025-08-28", 7.53), ("2025
    ("2025-09-01", 7.55), ("2025-09-02", 7.70), ("2025-09-03", 7.63), ("2025
    ("2025-09-05", 7.67), ("2025-09-08", 7.63), ("2025-09-09", 7.75), ("2025
    ("2025-09-11", 7.88), ("2025-09-12", 7.88), ("2025-09-15", 7.75), ("2025
    ("2025-09-17", 7.84), ("2025-09-18", 7.65), ("2025-09-19", 7.61), ("2025
    ("2025-09-23", 7.50), ("2025-09-24", 7.47), ("2025-09-25", 7.30), ("2025
    ("2025-09-29", 7.48), ("2025-09-30", 7.48), ("2025-10-02", 7.40), ("2025
    ("2025-10-06", 7.29), ("2025-10-08", 7.28), ("2025-10-09", 7.29), ("2025
    ("2025-10-13", 7.32), ("2025-10-14", 7.49), ("2025-10-15", 7.49), ("2025
    ("2025-10-17", 7.62), ("2025-10-20", 7.75), ("2025-10-21", 7.82), ("2025
    ("2025-10-23", 7.88), ("2025-10-24", 7.89), ("2025-10-27", 7.88), ("2025
    ("2025-10-30", 7.86), ("2025-10-31", 7.70), ("2025-11-03", 7.94), ("2025
    ("2025-11-05", 8.02), ("2025-11-06", 8.14), ("2025-11-07", 8.13), ("2025
    ("2025-11-11", 8.33), ("2025-11-12", 8.40), ("2025-11-13", 8.42), ("2025
    ("2025-11-17", 8.24), ("2025-11-18", 8.15), ("2025-11-19", 8.13), ("2025
    ("2025-11-21", 8.09), ("2025-11-24", 8.21), ("2025-11-25", 8.22), ("2025
    ("2025-11-27", 8.23), ("2025-11-28", 8.17), ("2025-12-01", 8.15), ("2025
    ("2025-12-03", 7.84), ("2025-12-04", 7.92), ("2025-12-05", 7.98), ("2025
    ("2025-12-09", 7.61), ("2025-12-10", 7.57), ("2025-12-11", 7.58), ("2025
    ("2025-12-15", 7.55), ("2025-12-16", 7.39), ("2025-12-17", 7.43), ("2025
    ("2025-12-19", 7.49), ("2025-12-22", 7.54), ("2025-12-23", 7.61), ("2025
    ("2025-12-29", 7.62), ("2025-12-30", 7.72),
]

dates = [datetime.strptime(d, "%Y-%m-%d") for d, _ in price_data]
prices = [p for _, p in price_data]
n_days = len(dates)

maturity_date = datetime.strptime("2025-12-30", "%Y-%m-%d")
start_date = datetime.strptime("2025-06-30", "%Y-%m-%d")

# Build hedging table
hedging_records = []

for i in range(n_days):
    date = dates[i]
    price = prices[i]
    T_remain = (maturity_date - date).days / 365.0
    num_days_from_prev = (dates[i] - dates[i - 1]).days if i > 0 else 0

    if i < n_days - 1:
        # Normal day: compute delta
        if T_remain > 0:
            d1_val = (np.log(price / K_option) + (r + 0.5 * sigma1**2) * T_r
            Nd1 = norm.cdf(d1_val)
        else:
            d1_val = np.nan
            Nd1 = 1.0 if price > K_option else 0.0
        delta_shares = Nd1 * M
    else:
        # Maturity day
        d1_val = np.nan
```

```
        if price > K_option:
            Nd1 = 1.0
            delta_shares = M
        else:
            Nd1 = 0.0
            delta_shares = 0

    if i == 0:
        shares_traded = delta_shares
        cost_of_trade = shares_traded * price
        interest = 0.0
        account_balance = -cost_of_trade
    else:
        prev = hedging_records[i - 1]
        interest = prev['Account Balance'] * (np.exp(r * num_days_from_prev
        shares_traded = delta_shares - prev['Delta Shares']
        cost_of_trade = shares_traded * price
        account_balance = prev['Account Balance'] + interest - cost_of_trade

    hedging_records.append({
        'Date': date.strftime('%Y-%m-%d'),
        'T_remain': round(T_remain, 4),
        'Price': price,
        'd1': round(d1_val, 4) if not np.isnan(d1_val) else np.nan,
        'N(d1)': round(Nd1, 4),
        'Delta Shares': round(delta_shares, 2),
        'Shares Traded': round(shares_traded, 2),
        'Cost of Trade': round(cost_of_trade, 2),
        'Interest': round(interest, 2),
        'Account Balance': round(account_balance, 2),
    })

df_hedge = pd.DataFrame(hedging_records)
print(f"Total trading days: {n_days}")
print(f"Final stock price on maturity (Dec 30): {prices[-1]}")
print(f"Strike price: {K_option}")
print(f"Option is {'IN' if prices[-1] > K_option else 'OUT-OF'}-THE-MONEY")
```

```
Total trading days: 126
Final stock price on maturity (Dec 30): 7.72
Strike price: 7.92
Option is OUT-OF-THE-MONEY
```

Hedging Table (Selected Rows)

First 10 trading days:

In [7]:
```
# First 10 rows
display(df_hedge.head(10))
```

	Date	T_remain	Price	d1	N(d1)	Delta Shares	Shares Traded	Cost of Trade	Interest
0	2025-06-30	0.5014	7.92	0.1816	0.5720	143011.21	143011.21	1132648.81	0.00
1	2025-07-02	0.4959	8.15	0.3818	0.6487	162176.91	19165.70	156200.45	-194.89
2	2025-07-03	0.4932	8.15	0.3819	0.6487	162182.35	5.44	44.35	-110.90
3	2025-07-04	0.4904	8.21	0.4338	0.6678	166946.40	4764.05	39112.84	-110.91
4	2025-07-07	0.4822	8.20	0.4258	0.6649	166215.03	-731.37	-5997.21	-342.89
5	2025-07-08	0.4795	8.24	0.4608	0.6775	169378.20	3163.17	26064.54	-113.80
6	2025-07-09	0.4767	8.22	0.4436	0.6713	167836.73	-1541.47	-12670.92	-116.05
7	2025-07-10	0.4740	8.48	0.6678	0.7479	186968.60	19131.87	162238.27	-114.97
8	2025-07-11	0.4712	8.35	0.5573	0.7113	177836.37	-9132.23	-76254.14	-128.94
9	2025-07-14	0.4630	8.41	0.6112	0.7295	182369.72	4533.35	38125.47	-367.20

In [8]:
```
# Last 10 rows (approaching maturity)
print("Last 10 trading days:")
display(df_hedge.tail(10))
```

```
Last 10 trading days:
```

	Date	T_remain	Price	d1	N(d1)	Delta Shares	Shares Traded	Cost of Trade	Interes
116	2025-12-15	0.0411	7.55	-1.1164	0.1321	33033.24	-23967.59	-180955.28	-145.4
117	2025-12-16	0.0384	7.39	-1.7006	0.0445	11127.95	-21905.29	-161880.09	-32.9
118	2025-12-17	0.0356	7.43	-1.6269	0.0519	12970.12	1842.17	13687.33	-18.9
119	2025-12-18	0.0329	7.52	-1.3685	0.0856	21395.94	8425.82	63362.17	-20.1
120	2025-12-19	0.0301	7.49	-1.5473	0.0609	15222.36	-6173.58	-46240.10	-25.6
121	2025-12-22	0.0219	7.54	-1.6062	0.0541	13529.16	-1693.20	-12766.72	-64.9
122	2025-12-23	0.0192	7.61	-1.3918	0.0820	20497.31	6968.15	53027.64	-20.5
123	2025-12-24	0.0164	7.56	-1.7633	0.0389	9730.40	-10766.91	-81397.81	-25.1
124	2025-12-29	0.0027	7.62	-3.6387	0.0001	34.25	-9696.15	-73884.68	-90.6
125	2025-12-30	0.0000	7.72	NaN	0.0000	0.00	-34.25	-264.41	-11.7

Settlement at Maturity & Final Account Balance (Fi)

In [9]:
```
# Settlement at maturity
final_stock_price = prices[-1]
final_balance = hedging_records[-1]['Account Balance']

print(f"Final stock price (Dec 30, 2025): {final_stock_price}")
print(f"Strike price K: {K_option}")
print()

if final_stock_price > K_option:
    print(f"★ Option is IN-THE-MONEY → Exercised")
    print(f"  Sell {M:,} shares at strike price K = {K_option}")
    exercise_cash = M * K_option
    Fi = final_balance + exercise_cash
    print(f"  Cash received from exercise = {M:,} × {K_option} = {exercise_c
else:
    print(f"★ Option is OUT-OF-THE-MONEY → NOT Exercised")
    remaining_shares = hedging_records[-1]['Delta Shares']
    sell_cash = remaining_shares * final_stock_price
    Fi = final_balance + sell_cash
    if remaining_shares > 0:
        print(f"  Sell remaining {remaining_shares:,.0f} shares at market pr
```

```
print(f"\n{'='*50}")
print(f"  Account balance before settlement: {final_balance:,.2f}")
print(f"  ⋆ Fi (final account balance) = {Fi:,.2f}")
print(f"{'='*50}")
```

```
Final stock price (Dec 30, 2025): 7.72
Strike price K: 7.92

⋆ Option is OUT-OF-THE-MONEY → NOT Exercised

==================================================
  Account balance before settlement: -136,576.84
  ⋆ Fi (final account balance) = -136,576.84
==================================================
```

Q3(ii): Option Premium Deposit & Comparison

On June 30, 2025, deposit the premium received from shorting M call options into a deposit account at r = 3.14% p.a. continuously compounded, maturing Dec 30, 2025.

$$P_i = \text{Premium} \times e^{rT}$$

In [10]:

```
# Q3(ii): Option Premium Deposit
T3 = (maturity_date - start_date).days / 365.0

# BS price for the call option with T = T3
d1_q3 = (np.log(S1_0 / K_option) + (r + 0.5 * sigma1**2) * T3) / (sigma1 * r
d2_q3 = d1_q3 - sigma1 * np.sqrt(T3)
bs_price_q3 = S1_0 * norm.cdf(d1_q3) - K_option * np.exp(-r * T3) * norm.cdf

# Total premium received from shorting M options
premium_received = bs_price_q3 * M

# Deposit at r for T3 years
Pi = premium_received * np.exp(r * T3)

print(f"T (June 30 to Dec 30) = {(maturity_date - start_date).days} days = {
print(f"BS call price = {bs_price_q3:.6f}")
print(f"Total premium received = {bs_price_q3:.6f} × {M:,} = {premium_receiv
print(f"Pi = {premium_received:,.2f} × exp({r} × {T3:.6f}) = {Pi:,.2f}")

print(f"\n{'='*60}")
print(f"  ⋆ Fi (hedging cost at maturity)     = {Fi:,.2f}")
print(f"  ⋆ Pi (deposit value at maturity)    = {Pi:,.2f}")
print(f"  |Fi| - Pi                           = {abs(Fi) - Pi:,.2f}")
print(f"  |Fi| / Pi                           = {abs(Fi)/Pi:.4f}")
print(f"{'='*60}")

# Summary table
summary = pd.DataFrame({
    'Item': ['Fi (hedging final balance)', 'Pi (deposit maturity value)', '|
    'Value': [f'{Fi:,.2f}', f'{Pi:,.2f}', f'{abs(Fi) - Pi:,.2f}', f'{abs(Fi)
})
display(summary.style.set_caption("Q3(ii): Comparison of Fi and Pi").hide(ax
```

```
T (June 30 to Dec 30) = 183 days = 0.501370 years
BS call price = 0.512573
Total premium received = 0.512573 × 250,000 = 128,143.37
Pi = 128,143.37 × exp(0.0314 × 0.501370) = 130,176.69

============================================================
  ★ Fi (hedging cost at maturity)     = -136,576.84
  ★ Pi (deposit value at maturity)    = 130,176.69
  |Fi| - Pi                           = 6,400.15
  |Fi| / Pi                           = 1.0492
============================================================
```

Q3(ii): Comparison of Fi and Pi

Item	Value
Fi (hedging final balance)	-136,576.84
Pi (deposit maturity value)	130,176.69
|Fi| - Pi	6,400.15
|Fi| / Pi	1.0492

Comment on Q3(ii)

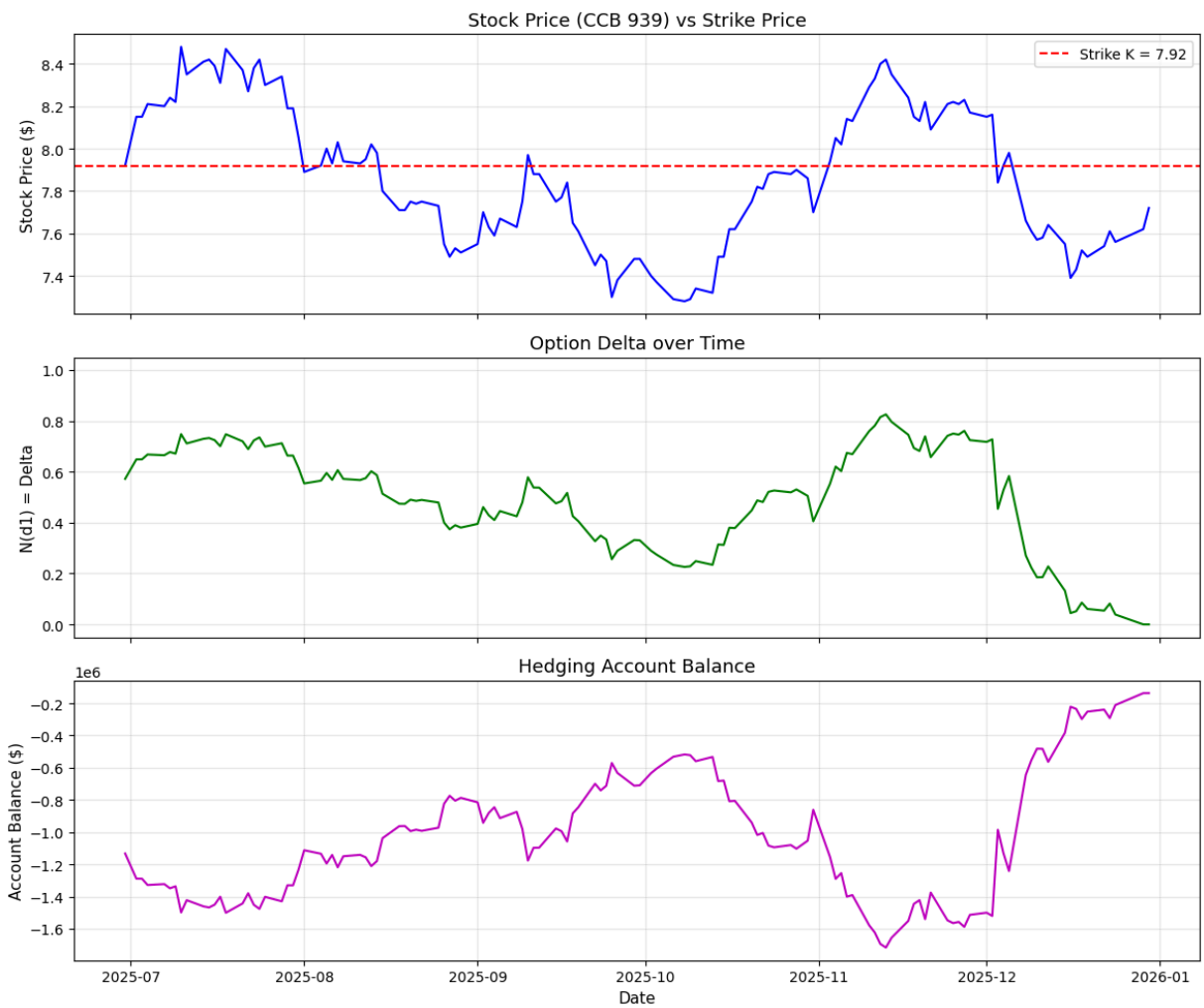
The final hedging account balance **Fi** is negative, while the deposit amount **Pi** is positive. Under the Black-Scholes framework, if delta hedging were performed **continuously**, |Fi| should exactly equal Pi — meaning the cost of hedging equals the option premium received.

In practice, since we hedge only at **discrete daily intervals**, there is a small discrepancy between |Fi| and Pi. This difference arises from:

1. **Discrete hedging error (gamma risk):** Rebalancing only once per day means the hedge is not perfect between rebalancing dates.
2. **Realized vs. implied volatility:** The actual stock price movements may exhibit different volatility from the implied volatility used for delta computation.

The closer |Fi| is to Pi, the more effective the delta hedging strategy is.

Visualization: Stock Price & Delta over Hedging Period

In [11]:
```
# Visualization
fig, axes = plt.subplots(3, 1, figsize=(12, 10), sharex=True)

plot_dates = [datetime.strptime(rec['Date'], '%Y-%m-%d') for rec in hedging_

# Stock price
axes[0].plot(plot_dates, [rec['Price'] for rec in hedging_records], 'b-', li
axes[0].axhline(y=K_option, color='r', linestyle='--', label=f'Strike K = {K
axes[0].set_ylabel('Stock Price ($)', fontsize=11)
axes[0].set_title('Stock Price (CCB 939) vs Strike Price', fontsize=13)
```

```
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# N(d1) = Delta
axes[1].plot(plot_dates, [rec['N(d1)'] for rec in hedging_records], 'g-', li
axes[1].set_ylabel('N(d1) = Delta', fontsize=11)
axes[1].set_title('Option Delta over Time', fontsize=13)
axes[1].set_ylim(-0.05, 1.05)
axes[1].grid(True, alpha=0.3)

# Account balance
axes[2].plot(plot_dates, [rec['Account Balance'] for rec in hedging_records]
axes[2].set_ylabel('Account Balance ($)', fontsize=11)
axes[2].set_title('Hedging Account Balance', fontsize=13)
axes[2].set_xlabel('Date', fontsize=11)
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```